

1

Ans to the que no: 1

Given,

$$\text{Mass, } m = 25\text{g} = 0.025\text{ kg}$$

$$\text{Spring constant, } k = 400\text{ dyn/cm} = 400\text{ Nm}^{-1}$$

$$\text{Displacement, } A = 10\text{cm} = 0.1\text{m}$$

①

$$\begin{aligned}\text{Time period, } T &= 2\pi\sqrt{\frac{m}{k}} \\ &= 2 \times 3.1416 \sqrt{\frac{0.025}{400}} \\ &= 0.0495\text{ s}\end{aligned}$$

②

$$\begin{aligned}\text{Frequency, } f &= \frac{1}{T} \\ &= \frac{1}{0.0495} = 20.20\text{ Hz}\end{aligned}$$

③

$$\begin{aligned}\text{Angular frequency, } \omega &= 2\pi f \\ &= 2 \times 3.1416 \times 20.20 \\ &= 127.2\text{ rad s}^{-1}\end{aligned}$$

(iv)

$$\begin{aligned}\text{Maximum velocity, } v_{\max} &= A\omega \\ &= 0.1 \times 127.2 \\ &= 12.72 \text{ ms}^{-1}\end{aligned}$$

Ans to the que no: 2

Given,

$$\text{Displacement, } x = 2.0 \text{ mm} = 0.002 \text{ m}$$

$$\text{Frequency, } f = 120 \text{ Hz}$$

(a)

$$\text{Amplitude, } A = \frac{x}{2} = \frac{0.002}{2} = 0.001 \text{ m}$$

(b)

$$\begin{aligned}\text{Maximum blade speed, } v_{\max} &= 2\pi f A \\ &= 2 \times 3.1416 \times 120 \times 0.001 \\ &= 0.754 \text{ ms}^{-1}\end{aligned}$$

(c)

$$\begin{aligned}\text{Magnitude of maximum acceleration, } a_{\max} &= \omega^2 A \\ &= (2\pi f)^2 A \\ &= (2 \times 3.1416 \times 120)^2 \times 0.001\end{aligned}$$

$$= 452.39 \text{ ms}^{-2}$$

Ans to the que no: 3

Given,

$$\text{Mass, } m = 0.12 \text{ kg}$$

$$\text{Amplitude, } A = 8.5 \text{ cm} = 0.085 \text{ m}$$

$$\text{Period, } T = 0.2 \text{ s}$$

$$\begin{aligned} \text{Angular frequency, } \omega &= \frac{2\pi}{T} = \frac{2 \times 3.1416}{0.2} \\ &= 10\pi \text{ rad s}^{-1} \end{aligned}$$

(a)

$$\begin{aligned} \text{Magnitude of maximum force, } f_{\text{max}} &= m\omega^2 A \\ &= 0.12 \times (10\pi)^2 \times 0.085 \\ &= 266.48 \text{ N} \end{aligned}$$

(b)

$$\begin{aligned} \text{Spring constant, } k &= m\omega^2 \\ &= 0.12 \times (10\pi)^2 \\ &= 1179.87 \text{ Nm}^{-1} \end{aligned}$$

Ans to the que no: 4

Given,

Mass of hydrogen atom, $m = 1.68 \times 10^{-27} \text{ kg}$

Frequency, $f = 10^{14} \text{ Hz}$

Amplitude, $A = 10^{-10} \text{ m}$

Angular frequency, $\omega = 2\pi f$
 $= 2\pi \times 10^{14} \text{ rad s}^{-1}$

$$\begin{aligned} \text{Force, } F &= m\omega^2 A \\ &= (1.68 \times 10^{-27}) \times (2\pi \times 10^{14})^2 \times 10^{-10} \\ &= 6.67 \times 10^{-9} \text{ N} \end{aligned}$$

Ans to the que no: 5

Given,

Velocity at mean position, $V_{\text{mean}} = 1 \text{ m s}^{-1}$

Acceleration at one extremity, $A_{\text{extremity}} = 1.57 \text{ ms}^{-2}$

Using these relationship we can determine the angular frequency, $\omega =$

$$1 = A\omega$$

$$\therefore A = \frac{1}{\omega}$$

and,

$$a_{\text{extremity}} = A\omega^2$$

$$\therefore 1.57 = A\omega^2$$

$$\Rightarrow 1.57 = \frac{1}{\omega} \times \omega^2$$

$$\therefore \omega = 1.57 \text{ rad s}^{-1}$$

$$\therefore \text{Time period, } T = \frac{2\pi}{\omega} = \frac{2 \times 3.1416}{1.57} \\ = 4 \text{ s}$$

Ans to the que no: 6

Given,

$$\text{Amplitude, } A = 5 \text{ m}$$

$$\text{Position, } x = 3 \text{ m from the mean position}$$

$$\text{Acceleration, } a = 48 \text{ ms}^{-2}$$

We know,

$$a = -\omega^2 x$$

6

$$\Rightarrow \omega^2 = \frac{a}{x}$$

$$\Rightarrow \omega = \sqrt{\frac{a}{x}}$$

$$\therefore \omega = \sqrt{\frac{48}{3}} = 16 \text{ rad s}^{-1}$$

(i)

Velocity, $v = \omega \sqrt{A^2 - x^2}$

$$= 4 \sqrt{5^2 - 3^2}$$
$$= 16 \text{ ms}^{-1}$$

(ii)

Time period, $T = \frac{2\pi}{\omega}$

$$= \frac{2 \times 3.1416}{4}$$
$$= 1.57 \text{ s}$$

(iii)

Maximum velocity, $v_{\text{max}} = \omega A$

$$= 4 \times 5 = 20 \text{ ms}^{-1}$$

Ans to the question:

Given,

Displacement at $t=0$, $x = 0.37\text{cm} = 0.0037\text{m}$

Frequency, $f = 0.25\text{Hz}$

(i)

$$\text{Time period, } T = \frac{1}{f} = \frac{1}{0.25} = 4\text{s}$$

(ii)

$$\begin{aligned}\text{Angular frequency, } \omega &= 2\pi f \\ &= 2 \times 3.1416 \times 0.25 \\ &= \frac{\pi}{2} \text{ rad}\end{aligned}$$

(iii)

$$\begin{aligned}\text{Amplitude, } A &= x_{\text{max}} \\ &= 0.0037\text{m}\end{aligned}$$

(iv)

Displacement calculator in Radian mode,

$$\begin{aligned}x &= A \cos(\omega t + \phi) \\ &= 0.0037 \cos\left(\frac{\pi}{2} \times 3 + 0\right) = 0\end{aligned}$$

⑤

$$\begin{aligned}
 v &= -A\omega \sin(\omega t + \phi) \\
 &= -0.003 \times \frac{\pi}{2} \sin\left(\frac{\pi}{2} \times 3 + 0\right) \\
 &= 5.809 \times 10^{-3} \text{ ms}^{-1}
 \end{aligned}$$

Ans to the que no: 8

Given,

Amplitude, $A = 4\text{m}$

Frequency, $f = 0.5\text{ Hz}$

phase angle, $\phi = \frac{\pi}{4}$

Angular frequency, $\omega = 2\pi f$
 $= 2\pi \times 0.5 = \pi \text{ rad s}^{-1}$

①

Time period, $T = \frac{1}{f} = \frac{1}{0.5} = 2.5$

②

Displacement, $x = A \sin(\omega t + \phi)$
 $= 4 \sin\left(\pi t + \frac{\pi}{4}\right)$

(iii)

$$x = 4 \sin\left(\pi t + \frac{\pi}{4}\right)$$

$$\therefore v = \frac{dx}{dt} = -4\pi \cos\left(\pi t + \frac{\pi}{4}\right)$$

$$t = 5s = -4\pi \cos\left(\pi \times 5 + \frac{\pi}{4}\right)$$

$$= -8.8857 \text{ ms}^{-1}$$

$$a = \frac{dv}{dt} = 4\pi^2 \sin\left(\pi t + \frac{\pi}{4}\right)$$

$$t = 5s = 4\pi^2 \sin\left(\pi \times 5 + \frac{\pi}{4}\right)$$

$$= 27.91 \text{ ms}^{-2}$$

Ans to the que no: 9

Given,

Mass, $m = 2 \text{ kg}$

Spring constant, $k = 196 \text{ Nm}^{-1}$

Amplitude, $A = 5 \text{ cm} = 0.05 \text{ m}$

Time period, $T = 0.8$

a

We know,

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{196}{2}} = 9.9 \text{ rad s}^{-1}$$

$$\therefore \omega = 2\pi f \Rightarrow f = \frac{\omega}{2\pi} = \frac{9.9}{2\pi} = 1.58 \text{ Hz}$$

$$\therefore T = \frac{1}{f} = \frac{1}{1.58} = 0.63 \text{ s}$$

b

Equation for x vs t ,

$$x = A \cos(\omega t + \phi)$$

$$\therefore x = 0.05 \cos(9.9t + 0)$$

$$= 0.05 \cos(9.9t)$$

Ans to the que no: 10

a

The time period of oscillation,

15 oscillation complete in 10 s

$$\therefore 1 \quad " \quad " \quad " \quad \frac{10}{15} \text{ s}$$

$$\therefore T = 0.67 \text{ s}$$

(b)

$$\text{Angular frequency, } \omega = \frac{2\pi}{T} = \frac{2\pi}{0.67} = 9.37 \text{ rad s}^{-1}$$

$$\therefore \text{Velocity, } V_{\max} = A\omega = 9.37 \times 0.2 = 1.874 \text{ ms}^{-1}$$

(c)

$$\text{Velocity, at } t = 0.8 \text{ s}$$

$$\text{Amplitude } A = 20 \text{ cm} = 0.2 \text{ m}$$

We know,

$$x = A \cos(\omega t)$$

$$= A \cos \frac{2\pi t}{T} = 0.2 \cos \frac{2\pi(0.8)}{0.67} = 0.19 \text{ m}$$

Ans to the que no. 11

$$\text{For, } R_{\max} = R$$

$$\frac{R^2}{4L^2} = \frac{1}{LC}$$

$$\Rightarrow \frac{R_{\max}^2}{4L^2} = \frac{1}{LC}$$

$$\begin{aligned}
 \therefore R_{\max} &= \sqrt{\frac{4L^2}{Lc}} \\
 &= \sqrt{\frac{4 \times (0.4)^2}{0.4 \times 0.002 \times 10^{-6}}} \\
 &= 284.27 \Omega
 \end{aligned}$$

$\therefore$ So that $R_{\max}$ is 284.27 or greater value to become oscillatory.

Ans to the que no: 12

angular frequency, $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{85}{0.25}} = 18.43$

We know,

$$T = \frac{2\pi}{\omega} = \frac{2 \times 3.1416}{18.43} = 0.3413 \text{ s} \quad \textcircled{a}$$

$\textcircled{b}$

We know,

$$\lambda = \frac{b}{2m} = \frac{0.07}{2 \times 0.25} = 0.14 \text{ rad s}^{-1}$$

$$t_{\text{half}} = \frac{\ln(2)}{\lambda} = \frac{\ln(2)}{0.14} = 4.96 \text{ s}$$

Ans to the que no: 13

(a)

Given,

$$\text{Mass, } m = (0.1 \times 56) = 5.6 \text{ kg}$$

$A(t) = 5$ s the $A(5)$ is equal to the half the initial $A \frac{A}{2}$

Amplitude of damping oscillator,

$$\frac{A_0}{2} = A_0 e^{-\gamma t}$$

$$\Rightarrow \frac{A_0}{2A_0} = e^{-\gamma t}$$

$$\Rightarrow \frac{1}{2} = e^{-\gamma t}$$

$$\ln\left(\frac{1}{2}\right) = -\gamma t$$

$$\therefore \gamma = \frac{\ln\left(\frac{1}{2}\right)}{-5} = 0.14$$

We know,

$$\gamma = \frac{b}{2m}$$

$$\begin{aligned} \therefore b &= \gamma \times 2m = 0.14 \times 2 \times 5.6 \\ &= 1.568 \text{ N s m}^{-1} \end{aligned}$$

(b)

$$\text{life time of oscillation} = \frac{1}{\gamma} = \frac{1}{\frac{b}{2m}} = \frac{1}{\frac{1.568}{5.6}} \\ = 3.57 \text{ oscillation}$$

Ans to the que no: 14

Given,

$$\text{Mass, } m = 1.5 \text{ kg}$$

$$\text{Spring constant, } k = 8 \text{ Nm}^{-1}$$

$$\text{Damping coefficient, } b = 230 \text{ gs}^{-1} = 0.23 \text{ kg s}^{-1}$$

$$\text{Displacement, } x = 12 \text{ cm} = 0.12 \text{ m}$$

$$n = 20$$

We know,

$$\omega' = \sqrt{\omega^2 - \gamma^2} = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2} = \sqrt{\frac{8}{1.5} - \left(\frac{0.23}{2 \times 1.5}\right)^2} \\ = 2.30 \text{ rad s}^{-1}$$

The differential equation of D.H.M,

$$x = A e^{-\frac{b}{2m}t} \cos(\omega't + \phi)$$

$$\Rightarrow 0.12 = A e^{-\frac{0.23}{2 \times 1.5} \times 0} \cos(2.30 \times 0 + \phi)$$

$$\Rightarrow 0.12 = A \cos 0$$

$$\therefore A = 0.12 \text{ m}$$

Now the rotation of 20 cycles,

$$\text{Rotation, } e^{-\frac{b}{2m}} \times n \times T$$

$$= e^{-\frac{0.23}{1 \times 1.5}} \times 20 \times 2.73$$

$$= 46.84 \text{ m}$$

$$\text{The ratio will be } = \frac{46.84}{0.12}$$

$$= \frac{1171}{3}$$

$$= 1171:3 \text{ (Ans)}$$